

SE LAB 1

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REQUIREMENTS TABLE:

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (short)	Comments
FR-001	Functional	The system shall allow authenticated students to search real-time availability and reserve a maximum 2-hour slot for hardware up to 7 days in advance.	High	Pass: Slot locks and token generates. Fail: Allows double-booking or exceeds 2 hours.	Prevents equipment scheduling conflicts	
FR-002	Functional	The system shall automatically verify equipment calibration status before confirming any slot reservation.	High	Pass: Reservation proceeds only if calibration flag is 'Valid'. Fail: Allows booking uncalibrated hardware.	Ensures safe and accurate experiment results	

FR-003	Functional	The system shall enable Lab Technicians to update equipment calibration records and flag hardware as 'Under Maintenance'.	Medium	Pass: Updating status immediately blocks booking for that unit. Fail: Hardware remains bookable during maintenance.	Keeps inventory and equipment health up to date	
FR-004	Functional	The system shall log check-in/check-out timestamps and calculate disciplinary late-return penalties automatically.	High	Pass: System applies fine/flag if check-out time > reservation end time. Fail: No penalty recorded for late returns.	Enforces discipline and timely equipment availability	
FR-005	Functional	The system shall send automated notification reminders to students 15 minutes before their reserved slot begins.	Low	Pass: Notification dispatched 15 mins prior to start time. Fail: Notification missing or sent after start time.	Reduces missed reservations and idle hardware	

NFR-001	Nonfunc.	The system shall process concurrent slot reservation requests within 200 ms with strict database-level lock isolation.	High	Pass: Benchmarking confirms target latency under peak load without race conditions. Fail: Response time > 200 ms or race condition occurs.	Prevents duplicate bookings under heavy load	
NFR-002	Nonfunc.	The system shall enforce role-based access control (RBAC) ensuring only authorized technicians can override reservation locks.	High	Pass: Student credentials denied when attempting technician overrides. Fail: Unauthorized user gains override access.	Maintains administrative integrity and security	

ACTORS:

Student: Reserves slots, checks in/out hardware, views notifications.

Lab Technician: Manages calibration, updates equipment status, overrides locks.

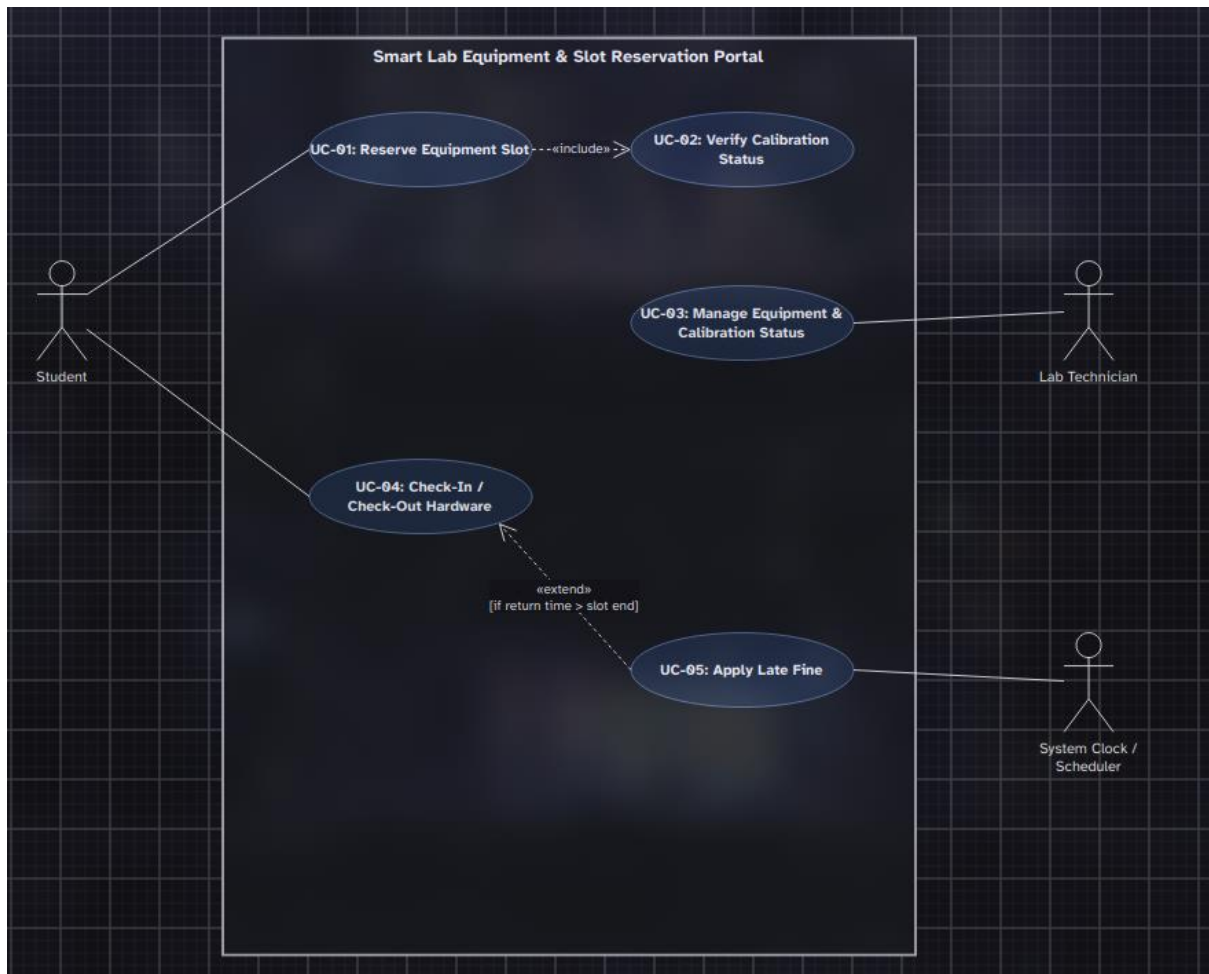
System Clock / Scheduler (Secondary): Triggers late penalties and automated reminders.

USE CASES:

- **UC-01:** Reserve Equipment Slot
- **UC-02:** Verify Calibration Status (include in UC-01)
- **UC-03:** Manage Equipment & Calibration (Lab Technician)
- **UC-04:** Check-In / Check-Out Hardware

- **UC-05: Apply Late Fine** (extend to UC-04 when return is overdue)

UML DIAGRAM



USE CASE FLOW SPECIFICATION:

Use Case ID: UC-01

Use Case Name: Reserve Equipment Slot

Primary Actor: Student

Preconditions: Student is logged in; requested hardware exists in inventory and is not flagged "Under Maintenance".

Postconditions: Success: Slot is locked, database is updated, and a unique reservation token is generated and issued to the student. Failure: No slot is locked and no reservation record is created.

Main Success Scenario:

1. Student selects desired equipment (e.g., Oscilloscope) and views real-time availability.
2. Student selects an available slot (maximum 2 hours) within 7 days.
3. System triggers UC-02: Verify Calibration Status.
4. System confirms equipment is calibrated and available.
5. System locks the time slot and issues a unique reservation token to the student.

Alternate Flow (A1: Equipment Uncalibrated):

- 3a. System detects equipment calibration status is 'Expired' or 'Pending'.
- 3b. System blocks reservation and displays message: "Selected hardware is currently uncalibrated and unavailable."
- 3c. System prompts student to select an alternate calibrated unit or time slot.

Use Case ID: UC-02

Use Case Name: Verify Calibration Status (included in UC-01)

Primary Actor: System (invoked on behalf of the Student during UC-01)

Preconditions: Student has selected equipment and a time slot in UC-01; a calibration record exists for the selected equipment.

Postconditions: Success: Calibration status "Valid" is returned to UC-01, which proceeds to lock the slot. Failure: Status "Expired", "Pending" or "Under Maintenance" is returned and UC-01 does not confirm the reservation. No equipment or calibration data is modified in either case.

Main Success Scenario:

1. System receives the equipment ID and requested slot from UC-01.
2. System retrieves the calibration record for that equipment.
3. System checks that the calibration flag is 'Valid' and the equipment is not flagged 'Under Maintenance'.
4. System returns the result "Calibrated" to UC-01, which continues with slot locking.

Alternate Flow (A1: Calibration Not Valid):

- 3a. System finds the calibration flag is 'Expired' or 'Pending', or the equipment is 'Under Maintenance'.
- 3b. System returns "Not Calibrated" to UC-01.
- 3c. UC-01 executes its alternate flow A1 and blocks the reservation.

Use Case ID: UC-03

Use Case Name: Manage Equipment & Calibration

Primary Actor: Lab Technician

Preconditions: Lab Technician is logged in with the Technician role (RBAC verified); the equipment exists in the inventory.

Postconditions: Success: The calibration record and status flag (Valid, Expired or Under Maintenance) are saved to the database. If the unit is set to 'Under Maintenance' it is immediately blocked from new bookings. Failure: The record is unchanged.

Main Success Scenario:

1. Lab Technician selects the equipment to be managed.
2. System displays the current calibration record and status.
3. Lab Technician enters the updated calibration date and sets the status flag (e.g., 'Valid' or 'Under Maintenance').
4. System verifies the Technician role and validates the entered data.
5. System saves the updated record and applies the new status immediately.
6. System displays a confirmation message to the Technician.

Alternate Flow (A1: Unauthorized User):

- 4a. System detects that the logged-in user does not hold the Technician role.
- 4b. System denies the update, displays "Access denied", and leaves the record unchanged.

Alternate Flow (A2: Invalid Data):

- 4a. System detects a missing or invalid calibration date or status.
- 4b. System rejects the update and prompts the Technician to correct the entry.

Use Case ID: UC-04

Use Case Name: Check-In / Check-Out Hardware

Primary Actor: Student

Preconditions: Student is logged in and holds a confirmed reservation with a valid reservation token; the current time is within the reserved slot window.

Postconditions: Success: Check-in and check-out timestamps are logged, the reservation is marked 'Completed', and the equipment is returned to 'Available'. If the return is overdue, UC-05 is triggered. Failure: No timestamp is logged and the equipment status is unchanged.

Main Success Scenario:

1. Student arrives at the lab and presents the reservation token.
2. System validates the token and confirms the current time is within the reserved slot.
3. System logs the check-in timestamp and marks the equipment 'In Use'.
4. Student uses the hardware and later returns it.
5. System logs the check-out timestamp.
6. System compares the check-out time with the reservation end time and finds the return is on time.
7. System marks the reservation 'Completed' and the equipment 'Available'.

Alternate Flow (A1: Invalid or Expired Token):

- 2a. System finds the token is invalid, already used, or outside the slot window.
- 2b. System denies check-in and displays an error message.

Extension Flow (A2: Overdue Return):

- 6a. System finds the check-out time is later than the reservation end time.
- 6b. System triggers UC-05: Apply Late Fine.

Use Case ID: UC-05

Use Case Name: Apply Late Fine (extends UC-04)

Primary Actor: System Clock / Scheduler (Secondary)

Preconditions: A reservation exists whose end time has passed, and the hardware check-out in UC-04 was logged after the reservation end time (or has not yet been logged).

Postconditions: Success: A fine/penalty record is created against the student's account, the reservation is flagged 'Late Return', and the student is notified. Failure: No penalty is recorded (only when the return was not late).

Main Success Scenario:

1. System Clock / Scheduler (or UC-04) detects that the check-out time is later than the reservation end time.
2. System calculates the overdue duration.
3. System computes the late penalty according to the lab policy.
4. System records the fine against the student's account and flags the reservation 'Late Return'.
5. System sends a notification to the student about the penalty.

Alternate Flow (A1: Return Not Late):

- 1a. System finds the check-out time is at or before the reservation end time.
- 1b. No penalty is applied and the use case ends.

Alternate Flow (A2: Hardware Not Yet Returned):

- 1a. Reservation end time has passed and no check-out has been logged.
- 1b. Scheduler flags the reservation as overdue and re-checks until check-out is recorded, then continues from step 2.